

Shipping and Fulfillment

What USSCOS needs to deliver — scanning, carriers, freight and returns · 23 September 2026

The aim. Every movement of stock and every step of an order is recorded as it happens, by the person doing it, on a device in their hand — so the count is right because the work made it right, not because someone reconciled it afterwards.

1. Where shipping stands today

Some of this is built and working. The gaps are specific rather than general.

Step	Status	Notes
Sales order created	Built	Created as draft — see the warning below
Reaches the shipping queue	Built	Queue groups orders by carrier
Pick and verify by scan	Built	Scan validates against the order, warns on wrong or over-picked item
Report short stock	Built	Note goes back against the order
Invoice the picked quantity	Built	Short picks invoice what shipped and leave the order open
Packing slip	Built	Prints today
Pack verification	Missing	No second check at the packing bench
Deduct stock on shipment	Missing	Shipping does not move inventory at all
Carrier label and tracking	Manual	Tracking typed in by hand
Freight quote and BOL	Manual	Kuebix quote retyped, BOL produced outside the system
Returns	Missing	Nothing exists — a main cause of the current drift

Worth checking first. A new sales order is created as **draft**, and the shipping queue only shows orders that are confirmed or later. If sales staff expect orders to flow straight to shipping, some may be sitting where the clerk cannot see them. This costs nothing to check and may explain orders that seem to go missing.

2. Scanning — and one decision that saves months

The device recommendation is the Zebra TC22: an Android mobile computer with an integrated scan engine, a readable screen, and a replaceable battery. That is a sound choice and worth following.

Where the plan can be simplified

The scanning document describes a dedicated warehouse application. That is the right end state, but it is not required to begin, and the difference in cost is large.

The TC22 runs Android with a full browser, and Zebra's DataWedge delivers a scan into whatever field has focus, exactly as a keyboard would. **The screens already built for USSCOS therefore run on it as they are.** The pick-and-verify screen was written for touch, holds focus for the scanner, and stores progress on the server so an interrupted job can be picked up by someone else.

	Web screens on the TC22	Dedicated Android app
Time to first use	Now — already built	Months
Works offline	No	Yes
Device management	Browser only	Full enterprise control
Cost to change	Deploy and it is live	App release each time
Hardware lock-in	None — any device with a browser	Per-platform build

Recommendation. Buy the TC22s and run the existing web screens on them. Prove the workflow with real staff and real stock first. Build a native application later only if offline working turns out to matter — and the honest test of that is whether Wi-Fi actually drops in the racks, which the pilot will answer.

Barcode coverage is the real constraint

From the current product file: **UPC on 38% of items, case barcodes on 49%**. So scanning covers about half the catalog and typing the SKU is the normal path for the rest, not an exception. Both are supported. Worth setting that expectation before the devices arrive, and worth adding barcodes to the fastest-moving items first rather than trying to reach all 765.

3. The shipping flow we are building toward

Stage	What happens	Built?
Confirm	Order moves from draft into the shipping queue	Yes
Pick	Scan each item against the order. Wrong or excess items are refused at the shelf	Yes
Short stock	Picker records what is missing, order is flagged and routed back	Yes
Pack	Second scan at the bench, package count, weight and dimensions captured	No
Rate	Freight quote pulled from Kuebix, or a FedEx rate for parcel	No
Label	Carrier label printed, tracking captured automatically	No
Ship	Stock deducted, order closed, invoice raised for what shipped	Part
Documents	Packing slip, and a bill of lading for freight	Part
Return	Goods back in, restock or scrap, credit raised	No

4. FedEx

What an integration gives you, in order of value:

- **Tracking captured automatically** instead of typed, which removes a transcription error from every parcel.
- **Labels printed from USSCOS**, so the address on the label is the address on the order by construction.
- **Rates at order time**, so shipping is quoted rather than estimated.
- **Delivery status back on the order**, which answers "where is it" without leaving the system.

Needed: a FedEx developer account, API credentials, and the account number. The work is self-contained and does not depend on anything else on this list.

This is an efficiency gain, not a blocker. Manual tracking entry works on day one. It should be scheduled after the inventory work, not before it.

5. Kuebix

Kuebix does publish an API, which makes automation realistic rather than hypothetical. Two endpoints matter:

Purpose	Endpoint
Request a fresh quote	POST /action/quickRate
Retrieve quotes already on a shipment	GET /shipments/{id}/rates/view

Quotes return carrier, total price, fuel surcharge, additional charges, transit days and a reference number. Authentication is an API username, API key and client ID.

Two questions for Kuebix before any work starts: is API access enabled on our subscription, and does it cost extra? Neither is answerable from their documentation, and both decide whether this is worth beginning.

How it would work

From the sales order, request or retrieve quotes, show the options with carrier, price and transit days, let the clerk choose, and store the chosen quote against the order. That removes the retyping and — more usefully — keeps the freight cost attached to the order instead of living on paper.

6. Freight and bills of lading

Worth building whether or not the Kuebix API is available. The API makes it quicker to fill in; it is not what makes it valuable.

- Carrier, service, quote reference and cost recorded against the order.
- Pallet count, weight and dimensions.
- **Bill of lading generated** rather than produced by hand.
- Freight cost visible against the order, so margin reflects what delivery actually cost.

7. Returns

Listed here because returns are a shipping activity and because they were named as a main cause of stock being wrong. Nothing exists for them today.

- What came back, from which customer, against which invoice.
- **A decision for each item: back into sellable stock, or scrapped.** These are different movements, and treating them the same rebuilds the drift.
- The stock movement that follows from that decision.
- A credit memo, or a record that one is owed.

The database already anticipates this — the invoice types include credit memo, and the stock movement types include return in and return out. The structure is there; the screens are not.

8. What is needed from USSC

Item	Why it matters	Who
Is Kuebix API access enabled on our subscription, and does it cost extra?	Decides whether freight is an integration or a data-entry screen	Kuebix support
FedEx account number and API credentials	Nothing can start without them	FedEx account admin
Product file with units per case, pallet quantity and barcodes	Scanning cannot convert a case into units without them	Whoever owns the spreadsheet
Does a pallet count as a quantity of cases, or its own item?	Decides how counting works across pack levels	Operations

Item	Why it matters	Who
Bays and racks in the 730 warehouse	Needed to record where stock actually sits	Operations
Is the draft-to-confirmed step happening on every order?	Orders may be invisible to shipping	Sales and shipping
Who prints labels, and on what printer?	Decides how label printing is wired	Shipping

The order that matters. Stock movements first — receiving, transfers, deduction on shipment, returns. Carrier and freight automation second. A perfectly automated shipping label on top of an inventory count nobody believes solves the smaller half of the problem.